Nuclear Astrophysics

Introductory guide

During the exercise sessions you will be asked to perform some data analysis tasks on a computer. Here is some helpful information.

- Each one of you will get an account to access the computers in room S201/3 (for the duration of the exercise sessions only). In addition, you get assigned a personal two-digit ID (01-30) which you will need to identify your working environment. Please keep this information safe, as you will need it for the duration of the semester.
- The exercise sheets can be downloaded from the MOODLE course webpage. This is also where you will have to hand in the exercises. For most exercises you will work in a JUPYTER NOTEBOOK (more information below), which you can simply upload on MOODLE. Handwritten calculations should be scanned and submitted in PDF format. The exercises are part of the evaluation to pass this course and you will have to submit at least 6 exercises from a total of 7.
- Many of the exercises will require the usage of software which is already installed on our computer cluster CCSN[1], such as JUPYTER NOTEBOOK and the nuclear reaction network code WINNET. In order to conveniently run them without having to install anything, we set up a way for you to access CCSN temporarily and take home the results. This requires you to follow these steps at the start of every exercise session:
  1. Choose a working computer in the computer room and log in with your account.
  2. In order to connect to your personal JUPYTER NOTEBOOK (which is running on CCSN), please open a terminal and type the following command (in one line):
     ```
     ssh -N -f -L 40XX:10.10.1.Y:40XX teaching@ccsn -J
         guest_arcones@theorie.ikp.physik.tu-darmstadt.de
     ```
     where "XX" has to be replaced by your personal two-digit ID and "Y" has to be replaced by "4" if your ID is between 01 and 16, and by "5" if it is 17 or higher.
  3. Call one of the teaching assistants to enter the required passwords.
  4. Open a browser and type in the address bar:
     ```
     localhost:40XX
     ```
     where again "XX" has to be replaced by your ID. If everything worked, you will see a JUPYTER login prompt. Click on "log in" without typing a password. JUPYTER LAB should open. It is running in your personal working folder on CCSN, where you can store your files.
- If you prefer to use your own device for the data analysis, you can download the ANACONDA package that includes JUPYTER NOTEBOOK from `https://www.anaconda.com/products/distribution`. You may also have to download the necessary data and notebooks from CCSN (see below). If you are interested in running WINNET on your own device, you can download it from `https://github.com/nuc-astro/WinNet`, although it requires a working Fortran compiler.

[1] CCSN = **C**omputational **C**luster for **S**imulations in **N**uclear astrophysics. Any resemblance to another commonly used acronym in nuclear astrophysics is highly intentional.

- Each Jupyter notebook consists of cells, which can contain either text and equations in markdown syntax or Python code. In order to run a selected cell, use `Ctrl+Enter`. You can read more about the basics of Jupyter notebook at `https://jupyter-notebook.readthedocs.io/en/stable/examples/Notebook/Notebook%20Basics.html`.

- In order to take files home with you, you have to first copy them from your work directory on CCSN to the local computer. This can be achieved with the `scp` command. If you want to copy your whole work directory `workdir_X` (again "X" is your ID - however, here IDs between 1 and 9 are only 1-digit without a leading 0) to the current directory ("." - the dot at the end is part of the commands!) of your local computer, type in a terminal:

  ```
  scp -r teaching@ccsn:/home/teaching/nucastro_ex/workdir_X/ .
  ```

  Likewise, if you only want to copy a certain sub folder, use:

  ```
  scp -r teaching@ccsn:/home/teaching/nucastro_ex/workdir_X/path/to/subfolder/ .
  ```

  In case you only want a specific file, use:

  ```
  scp teaching@ccsn:/home/teaching/nucastro_ex/workdir_X/path/to/file.dat .
  ```

  In each case you will be asked to give a password, which has to be entered by one of the teaching assistants.